# Power Flow Analysis: Methodology and Applications

## Fundamentals of Electrical Power Systems - Week 8


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# Outline

# Power Flow Analysis Definition

## Definition

**Power flow analysis** (load flow analysis) is the calculation to determine the steady-state operating conditions of an electrical power system.

**Analysis calculates:**

- Voltage magnitude and phase angle at each bus
- Active and reactive power flow on each transmission line/transformer
- Power losses in the system
- Required generation

# Why Power Flow Analysis Matters

**Fundamental Importance:**

- Most fundamental tool in power systems study
- Basis for all advanced analyses
- Economic dispatch
- System expansion planning

**Advanced Applications:**

- Stability studies
- Contingency analysis
- Optimal power flow (OPF)
- Renewable integration studies

**Key Point**

Power flow analysis is the **foundation** for all power system studies!

# Power Flow Applications in Planning

**Planning Applications:**

1. **System Expansion Evaluation**
   - Determine need for additional generation
   - Evaluate new transmission line additions
   - Optimal location analysis for new equipment
2. **Contingency Studies (N-1, N-2)**
   - Impact analysis of component loss
   - Critical component identification
3. **Renewable Energy Integration**
   - Variability impact assessment
   - Reactive power compensation needs

# Power Flow Applications in Operations

**Operational Applications:**

1. **Economic Dispatch**
   - Optimal generation allocation based on cost
   - Considering power flow constraints
2. **Voltage Control**
   - Transformer tap setting determination
   - Capacitor bank placement and sizing
3. **Overload Detection**
   - Identify overloaded lines/transformers
   - Network reconfiguration to avoid overload

# Bus Model in Power Systems

**Each bus has 4 variables:**

1. $|V_i|$: Voltage magnitude at bus $i$ (pu or kV)
2. $\delta_i$: Voltage phase angle at bus $i$ (degrees or radians)
3. $P_i$: Active power injection at bus $i$ (MW or pu)
4. $Q_i$: Reactive power injection at bus $i$ (MVAr or pu)

**Sign convention:**

- $P_i > 0$: Bus is generator (injects power to system)
- $P_i < 0$: Bus is load (absorbs power from system)
- $Q_i > 0$: Reactive power injection (leading/capacitive)
- $Q_i < 0$: Reactive power absorption (lagging/inductive)

# Bus Classification

**Three types of buses based on known and unknown variables:**

**Slack Bus (Swing/Reference)**

- Known: $|V|$, $\delta$
- Unknown: $P$, $Q$

**PV Bus (Generator)**

- Known: $P$, $|V|$
- Unknown: $Q$, $\delta$

**PQ Bus (Load)**

- Known: $P$, $Q$
- Unknown: $|V|$, $\delta$

**Important**

Slack bus provides reference angle and balances system losses!

# Y-Bus Matrix Properties

**Bus Admittance Matrix:** $[I_{bus}] = [Y_{bus}][V_{bus}]$

**Properties:**

- **Symmetric:** $Y_{ik} = Y_{ki}$ (for networks without phase shifters)
- **Sparse:** Mostly zero elements due to limited interconnections
- **Singular:** No reference ground without slack bus
- **Diagonally dominant:** Diagonal elements dominate row sums

**Design Tip**

Sparse matrix techniques reduce computational complexity!

# Newton-Raphson Method Overview

**Core Concept:** Uses Taylor series linearization to solve non-linear power flow equations

**For function** $f(x) = 0$**:** $x^{(k+1)} = x^{(k)} - \frac{f(x^{(k)})}{f'(x^{(k)})}$

**For multi-variable systems:** $\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} - [J(\mathbf{x}^{(k)})]^{-1}\mathbf{f}(\mathbf{x}^{(k)})$

## Key Advantages

- Quadratic convergence (very fast: 3-5 iterations)
- Not sensitive to system size
- Robust for well-conditioned systems

# Newton-Raphson Power Flow Implementation

**Power Mismatch Calculation:**

- $\Delta P_i^{(k)} = P_i^{spec} - P_i^{calc}(V^{(k)}, \delta^{(k)})$
- $\Delta Q_i^{(k)} = Q_i^{spec} - Q_i^{calc}(V^{(k)}, \delta^{(k)})$

**Linearized Equation:**

$$\begin{bmatrix} \Delta P \\ \Delta Q \end{bmatrix} = \begin{bmatrix} \frac{\partial P}{\partial \delta} & \frac{\partial P}{\partial |V|} \\ \frac{\partial Q}{\partial \delta} & \frac{\partial Q}{\partial |V|} \end{bmatrix} \begin{bmatrix} \Delta \delta \\ \Delta |V| \end{bmatrix}$$

**Convergence Check**

Continue iterations until $\max(|\Delta P|, |\Delta Q|) < \epsilon$

# Gauss-Seidel Method Overview

**Core Concept:** Iterative method calculating variables sequentially using latest values

**Based on:** $V_i = \frac{1}{Y_{ii}} \left[ \frac{P_i - jQ_i}{V_i^*} - \sum_{k=1,k\neq i}^{n} Y_{ik} V_k \right]$

**For PQ buses:** $V_i^{(k+1)} = \frac{1}{Y_{ii}} \left[ \frac{P_i - jQ_i}{(V_i^{(k)})^*} - \sum_{j=1}^{i-1} Y_{ij} V_j^{(k+1)} - \sum_{j=i+1}^{n} Y_{ij} V_j^{(k)} \right]$

## Advantages

- Simple implementation
- Low memory requirements
- Good for learning/teaching

# Method Comparison

| Aspect | Newton-Raphson | Gauss-Seidel |
|---|---|---|
| Number of iterations | 3-5 | 20-100+ |
| Convergence type | Quadratic | Linear |
| Complexity per iter | $\mathcal{O}(n^{1.5-3})$ | $\mathcal{O}(n^2)$ |
| Total computation | Fast | Slow |
| Memory requirement | Large | Small |
| Implementation | Complex | Simple |
| Large systems (¿100) | Excellent | Poor |

Table: Newton-Raphson vs Gauss-Seidel Comparison

**Recommendation:** Newton-Raphson for all practical applications!

# MATPOWER Introduction

**Definition**

MATPOWER is an open-source package for power system simulation and optimization.

**Key Features:**

- Power flow (Newton, Gauss-Seidel, Fast-Decoupled, DC)
- Optimal Power Flow (OPF)
- Continuation Power Flow (CPF) for voltage stability
- Library of test systems (IEEE 9, 14, 30, 57, 118, 300 bus, etc.)
- Compatible with MATLAB and GNU Octave

**Website:** `https://matpower.org`

# MATPOWER Installation

- `% Download MATPOWER from https://matpower.org`
- `% Extract to folder, e.g.,: ~/matpower8.0`
- `% Add to Octave path`
- `addpath('~/matpower8.0');`
- `% Test installation`
- `mpver`

## Structure Data

MATPOWER uses `mpc` (MATPOWER case) structure for all system information:

- `mpc.baseMVA` - Base power
- `mpc.bus` - Bus data
- `mpc.gen` - Generator data
- `mpc.branch` - Branch data

# Running Power Flow in MATPOWER

- `% Load IEEE 9-bus system`
- `mpc = loadcase('case9');`
- `% Run power flow (default:  Newton-Raphson)`
- `results = runpf(mpc);`
- `% Display results`
- `printpf(results);`
- `% Access specific data`
- `V_mag = results.bus(:,8); % Voltage magnitude`
- `V_ang = results.bus(:,9); % Voltage angle`
- `P_gen = results.gen(:,2); % Generator real power`

**Pro Tip**

MATPOWER is the industry standard for power flow research and teaching!

# Power Flow Problem Solving Steps

**Step-by-Step Process:**

1. **System Data Preparation**
   - Define buses (Slack, PV, PQ)
   - Specify loads and generation
   - Define transmission lines/transformers
2. **Power Flow Solution**
   - Choose appropriate method
   - Set convergence criteria
   - Execute solution algorithm
3. **Result Analysis**
   - Check voltage limits
   - Verify line loading
   - Assess system losses

# Common Power Flow Problems

**Voltage Issues:**

- **Under Voltage:** $|V_i| < 0.95$ pu
  - Add reactive compensation
  - Adjust transformer taps
  - Increase generation voltage
- **Over Voltage:** $|V_i| > 1.05$ pu
  - Reduce reactive compensation
  - Add shunt reactors
  - Lower generator voltage

**Loading Issues:** Overloaded lines/transformers

- Redispatch generation
- Reconfigure network topology
- Add parallel lines

# Problem Set: Power Flow Analysis

**C1 (Remember):** What are the three types of buses in power flow analysis?

**C2 (Understand):** Explain the difference between Newton-Raphson and Gauss-Seidel methods.

**C3 (Apply):** For a 3-bus system with bus 1 as slack ($1.0\angle 0^\circ$), bus 2 as PV (1.05 pu, 200 MW), and bus 3 as PQ (100+j50 MVA), identify known and unknown variables.

**C4 (Analyze):** Compare computational efficiency of both methods for IEEE 118-bus system.

**C5 (Evaluate):** Evaluate the importance of power flow analysis in modern grid with renewable integration.

**C6 (Create):** Design a step-by-step tutorial for implementing Newton-Raphson in MATLAB/Octave.

# Real-World Case Study

## Scenario: Sumatra Grid Stability

A power system engineer in Sumatra needs to analyze power flows after adding renewable generation.

**Using Power Flow Analysis:**

1. **Before Integration:** Baseline system performance
2. **Renewable Addition:** PV and wind farm integration
3. **Voltage Impact:** Assess voltage rise from distributed generation
4. **Stability Assessment:** Determine system margins

*Power flow analysis helps make data-driven decisions for grid integration!*

# Key Takeaways

## What You Learned

- Power flow analysis fundamentals
- Three bus types: Slack, PV, PQ
- Newton-Raphson and Gauss-Seidel methods
- MATPOWER applications
- Problem identification and solutions

## Skills Developed

- Power system problem-solving
- Computational thinking
- Technical analysis
- Practical application of theory

# Next Steps

**Continue Your Learning:**

1. Practice power flow calculations with MATPOWER
2. Explore optimal power flow concepts
3. Study voltage stability analysis
4. Experiment with renewable integration scenarios
5. Join power systems community forums

**Resources:**

- MATPOWER User's Manual: `https://matpower.org/docs/manual.pdf`
- Power Systems Analysis Texts
- IEEE Test Systems: `https://labs.ece.uw.edu/pstca/`

# Questions & Discussion

# Thank You!

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